Skip Module Hyperparameter Selection

1 Algorithm

Algorithm 1 Skip Module Hyperparameter Selection

Require: parameter space Θ, validation set D_{val}, detector M, recall tolerance δ_R, time budget β, weights w_S, w_F, w_R

Ensure: optimal parameters θ^*

```
baseline ← evaluate_pipeline(D_val, skip=None, M)
extract R_base, FAR_base, T_DL from baseline
best_score ← −∞
best_θ ← None
for θ ∈ Θ do
    skip ← SkipModule(θ)
    results ← evaluate_pipeline(D_val, skip, M)
    extract R(θ), FAR(θ), S(θ), T_skip(θ) from results
    if R(θ) ≥ R_base − δ_R  and  T_skip(θ) ≤ β · T_DL then
        ΔFAR~(θ) ← max(0, (FAR_base − FAR(θ)) / FAR_base)
        R~(θ) ← 1 − (R_base − R(θ)) / δ_R
        score ← w_S S(θ) + w_F ΔFAR~(θ) + w_R R~(θ)
        if score > best_score then
            best_score ← score
            best_θ ← θ
        end if
    end if
end for
return best_θ
```

where, in the notation of the algorithm:

$$\widetilde{\Delta\mathrm{FAR}}(\theta) \leftarrow \max\left(0, \frac{\mathrm{FAR}_{\text{base}} - \mathrm{FAR}(\theta)}{\mathrm{FAR}_{\text{base}}}\right)$$

$$\widetilde{R}(\theta) \leftarrow 1 - \frac{R_{\text{base}} - R(\theta)}{\delta_R}$$

$$\text{score} \leftarrow w_S S(\theta) + w_F \widetilde{\Delta\mathrm{FAR}}(\theta) + w_R \widetilde{R}(\theta)$$